

we print a friendly message telling the user how to avoid this kind of error. The program continues to run, and the user never sees a traceback:

```
Give me two numbers, and I'll divide them.
Enter 'q' to quit.
```

```
First number: 5
Second number: 0
You can't divide by 0!
```

```
First number: 5
Second number: 2
2.5
```

```
First number: q
```

The only code that should go in a try block is code that might cause an exception to be raised. Sometimes you'll have additional code that should run only if the try block was successful; this code goes in the else block. The except block tells Python what to do in case a certain exception arises when it tries to run the code in the try block.

By anticipating likely sources of errors, you can write robust programs that continue to run even when they encounter invalid data and missing resources. Your code will be resistant to innocent user mistakes and malicious attacks.

Handling the FileNotFoundError Exception

One common issue when working with files is handling missing files. The file you're looking for might be in a different location, the filename might be misspelled, or the file might not exist at all. You can handle all of these situations with a try-except block.

Let's try to read a file that doesn't exist. The following program tries to read in the contents of *Alice in Wonderland*, but I haven't saved the file *alice.txt* in the same directory as *alice.py*:

```
alice.py from pathlib import Path

path = Path('alice.txt')
contents = path.read_text(encoding='utf-8')
```

Note that we're using `read_text()` in a slightly different way here than what you saw earlier. The encoding argument is needed when your system's default encoding doesn't match the encoding of the file that's being read. This is most likely to happen when reading from a file that wasn't created on your system.

Python can't read from a missing file, so it raises an exception:

```
Traceback (most recent call last):
❶ File "alice.py", line 4, in <module>
❷   contents = path.read_text(encoding='utf-8')
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```